

TFET Transistor Emulation with CMOS Technology

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Introduction

At present, MOSFETs are the most used transistors at technological level. The TFET transistors are a new range which is still under investigation and consideration to be used as substitute transistors of MOSFETs. The TFET transistors have a similar manufacturing to the MOSFET transistors, although a different operation. [2]

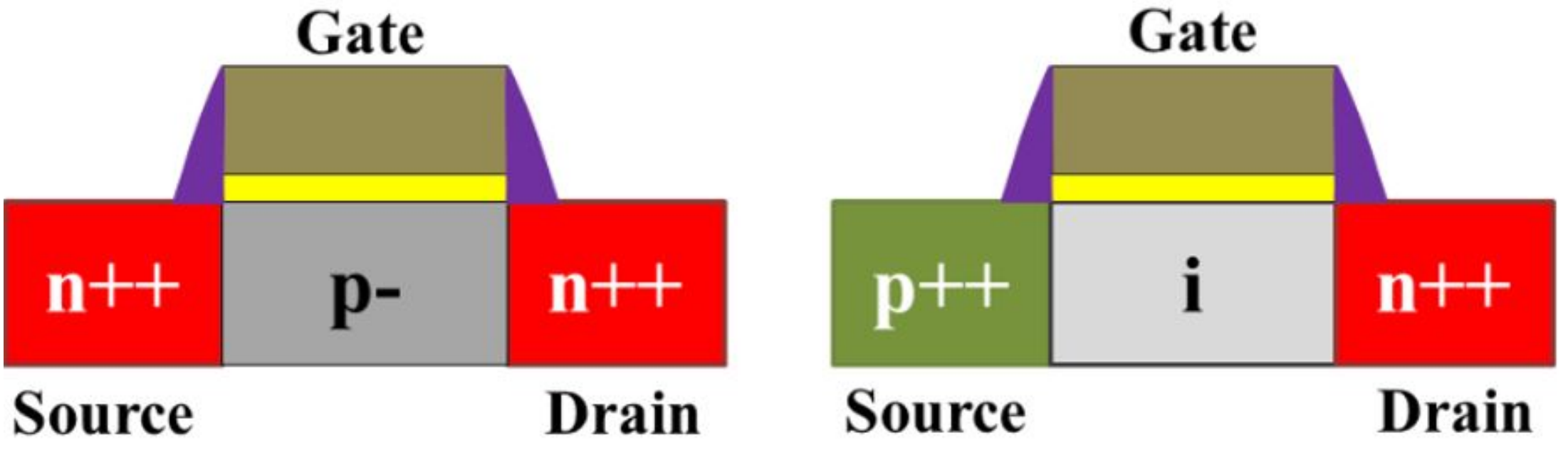


Figure 1: MOSFET transistor on the left and TFET on the right.

MOSFET transistors are characterized by allowing conduction due to a channel created by electrons between their Drain and Source terminal (Figure 2), this channel is generated due to voltage differential between the Gate and Bulk terminals.

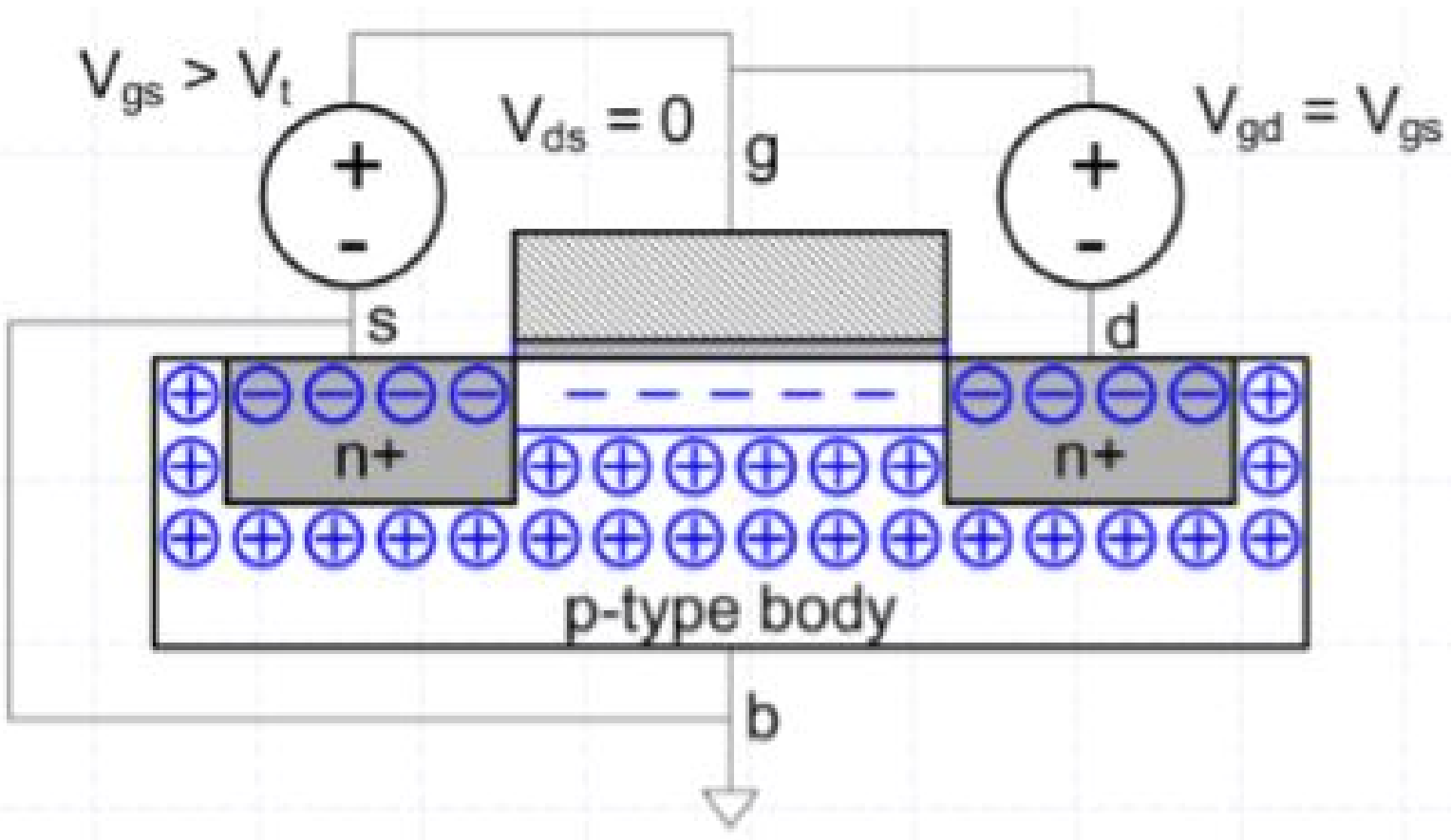


Figure 2: Electron channel formed in the MOSFET transistor.

At the time of their manufacture, TFET transistor get a physical channel which widens or narrows depending on the voltage difference between the Gate and Bulk gates (Figure 3).[1]

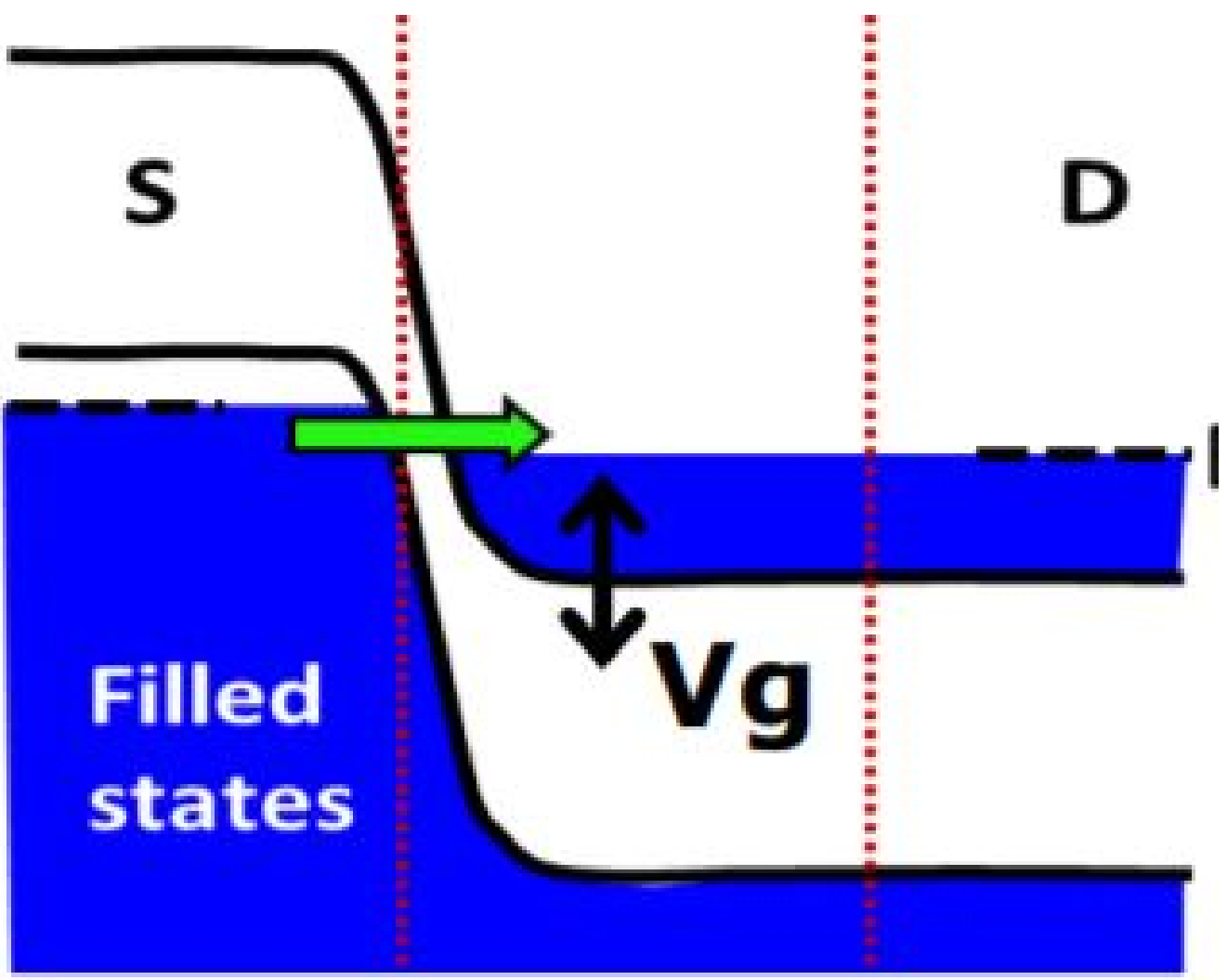


Figure 3: TFET Transistor channel.

TFET transistor channel requires lower voltage for its operation (Figure 4). [3] This means a decrease in energy consumption, what can be translated as an increase in batteries or electronics life.

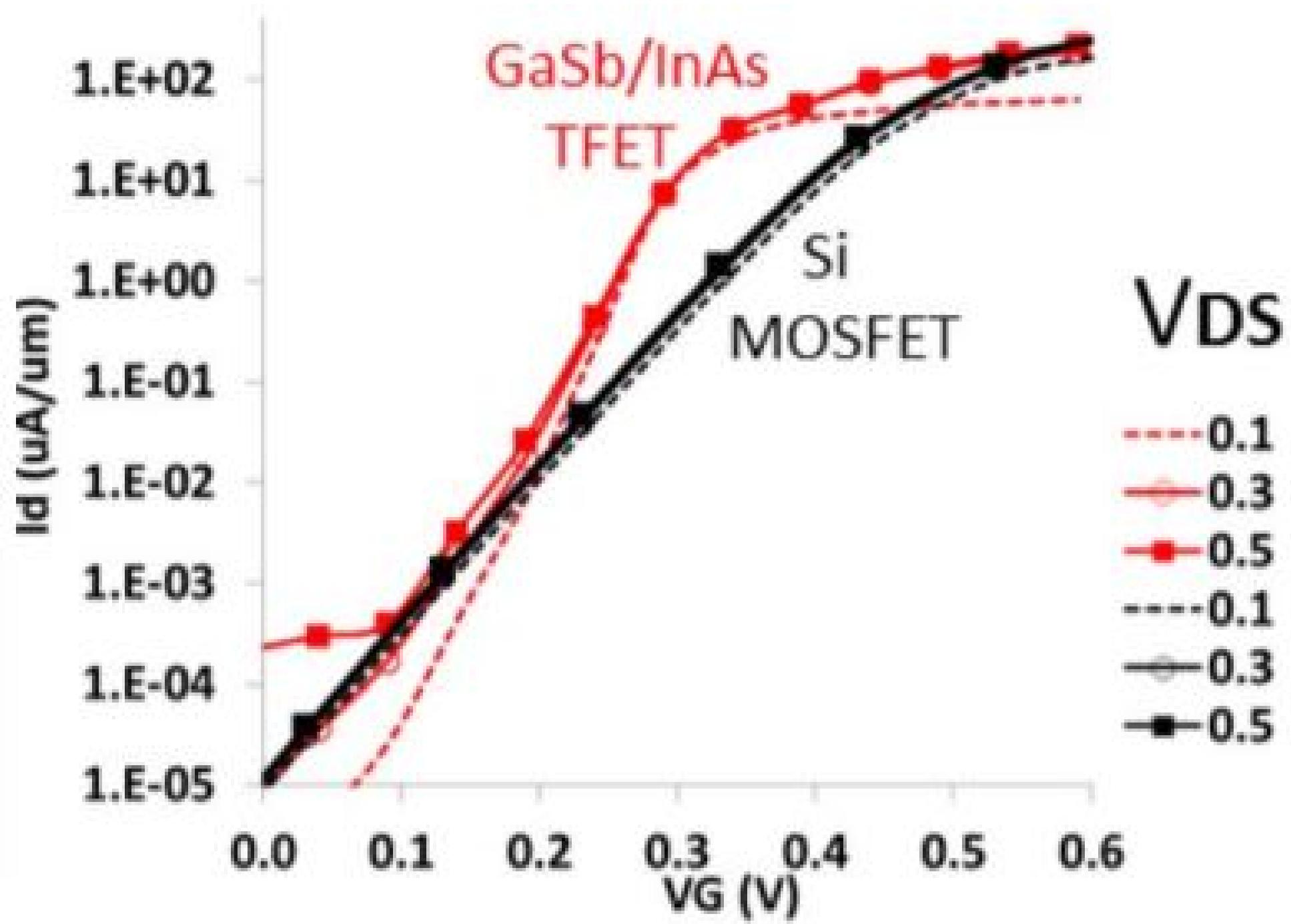


Figure 4: Comparison between MOSFET and TFET transistors.

Objectives

Emulate a TFET transistor response using CMOS processing simulation to probe if it's possible to replicate a futuristic technology with the technology used nowadays.

Methods

To accomplish the objectives, the HSPICE tool was used. This tool is used by manufacturers such as Intel for the design and study of transistors. This was provided by the Universidad del Valle de Guatemala.

As a starting point, we try to work with a single n-type MOSFET transistor, modifying its characteristics to generate a behavior similar to the TFET type transistor. Looking for a curve similar to the one shown in figure 4. The process was repeated for a p-type MOSFET transistor.

The modified n-type and p-type MOSFET transistors were taken and a not gate was made with them. Subsequently a pulse type signal was placed at the entrance of the gates, both modified and unchanged.

The frequency of this pulse was increased until obtaining the maximum frequency at which it was possible to operate for both gates. Finally, the power levels for the modified individual transistors and the modified gate were obtained. As well as for unmodified transistors and gate.

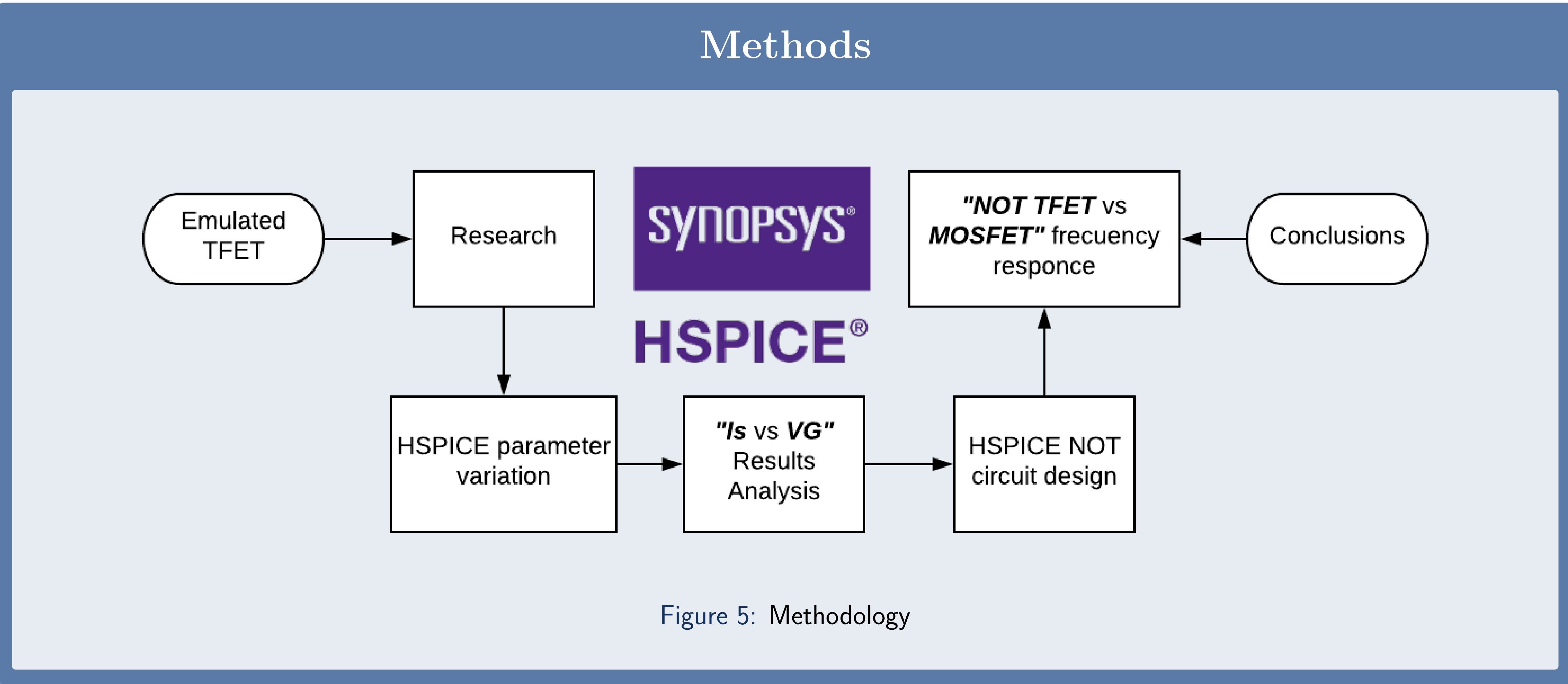


Figure 5: Methodology

Results

Table 1: Transistor Power Consumption

Component	Static (nW)	Dinamic (nW)
MOSFET	36411	10624
Emulated TFET	59510	32318
P-MOSFET	0.216	120.63
Emulated P-TFET	1.004	74.516
NOT	46.034	5475
TFET NOT	4652.9	20512

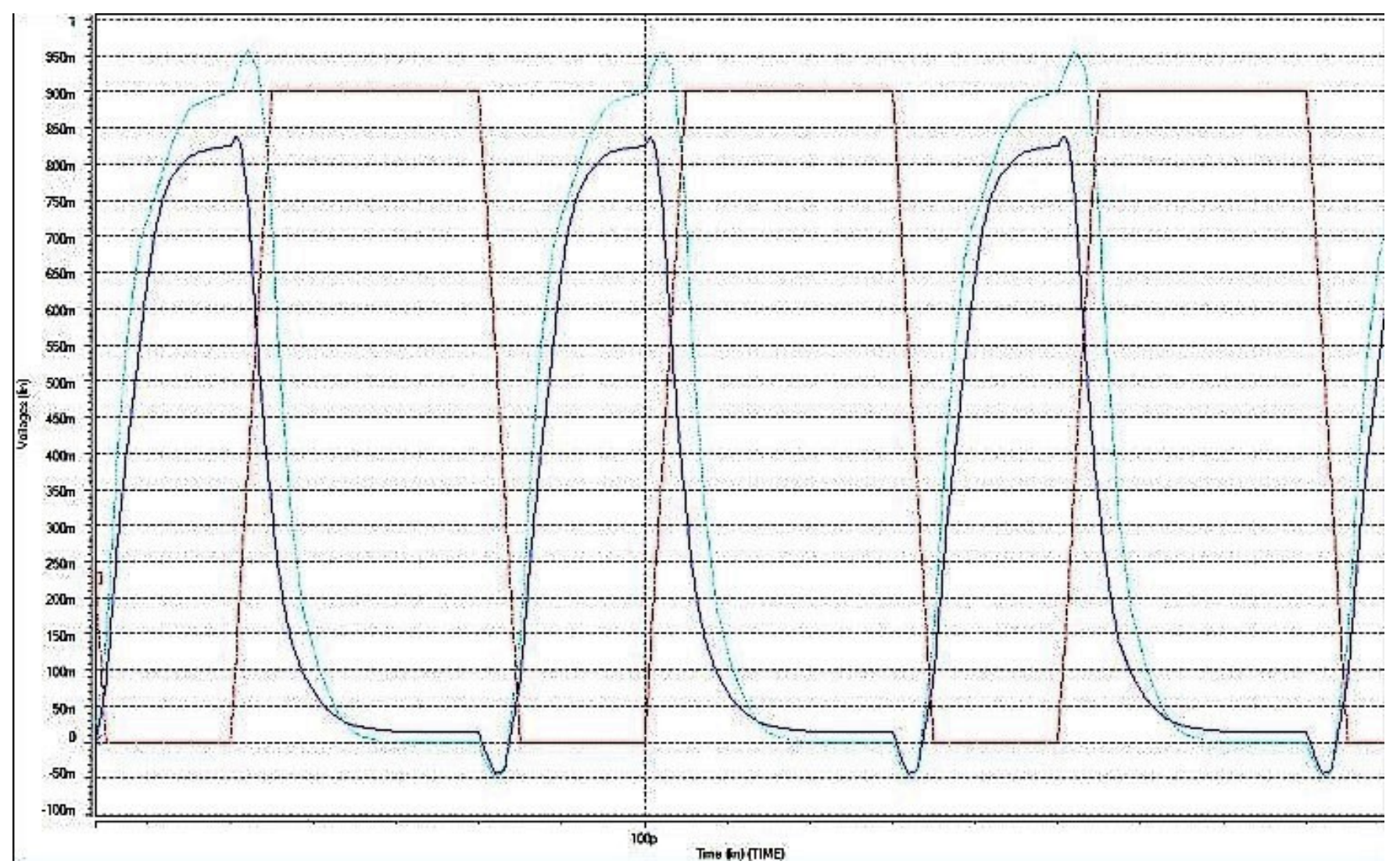


Figure 8: Maximum operating frequency for NOT gates.

Conclusion and Future Work

- A less gate voltage and same source current response, similar to that generated by a TFET transistor, was obtained in a n-type modify MOSFET transistor.
- A less gate voltage and less source current response was obtained in a p-type modify MOSFET transistor.
- In most cases a greater power was obtained in Emulated TFET transistors and gate.
- Maximum frequency for both modified and unmodified gates was 20 GHz.
- Not gate elaborates with unmodified transistors presented a faster low to high response.
- Not gate elaborates with emulated TFET transistors presented a faster high to low response.
- In future work, we hope to determine if it is possible to obtain a better result in the areas of energy consumption and frequency, as well as if it is possible to work with emulated TFET transistors in larger circuits as adders, and verify if this provides an improvement.

Referencias

- [1] Nikonov Dmitri. Tunneling FETs. INTEL.
- [2] K. Sivani. Modelling of Tunnel Field-Effect Transistor for Ultra-low-power Applications. Research Gate 2018.

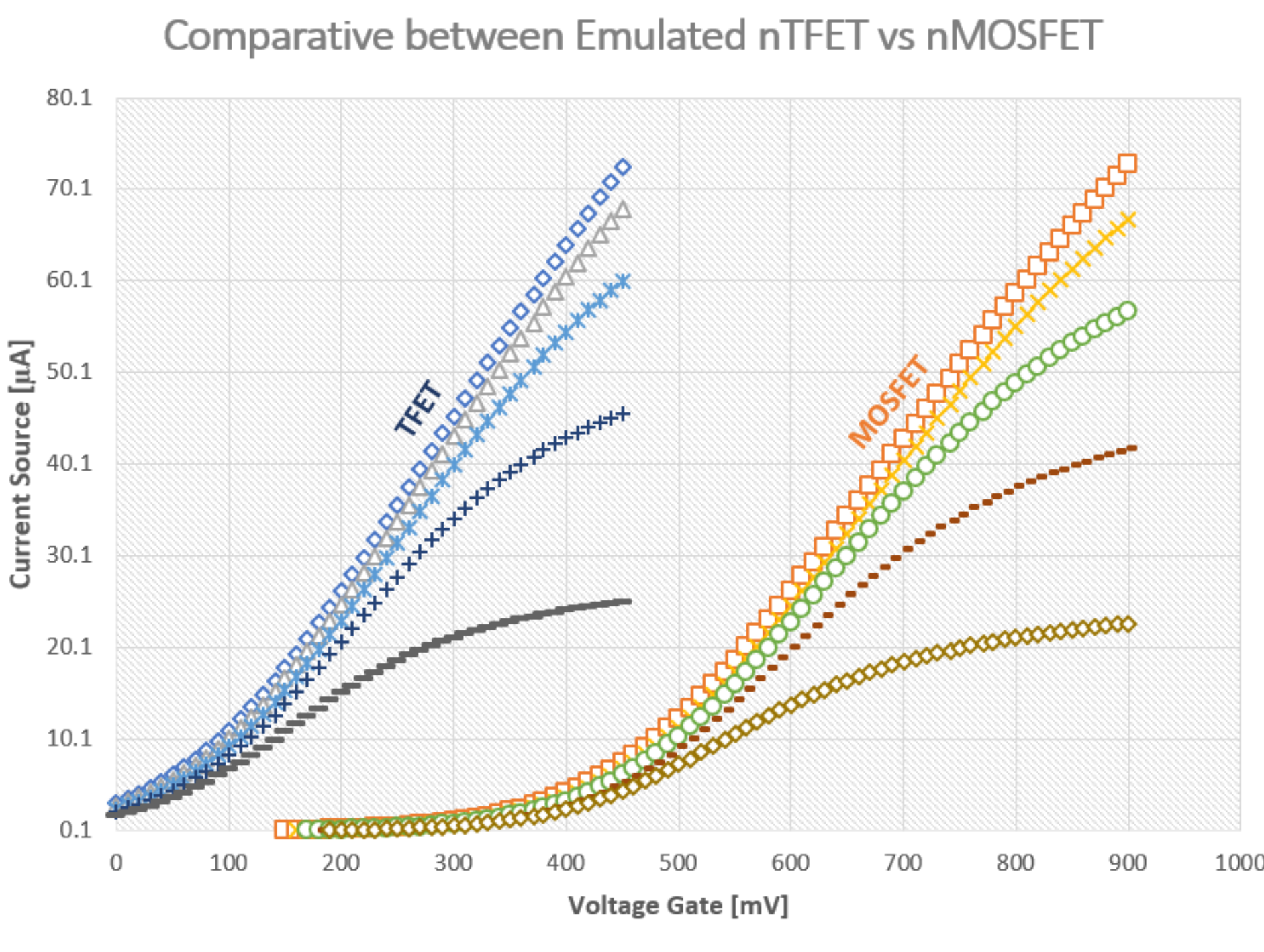


Figure 6: NMOSFET and NTFET Results.

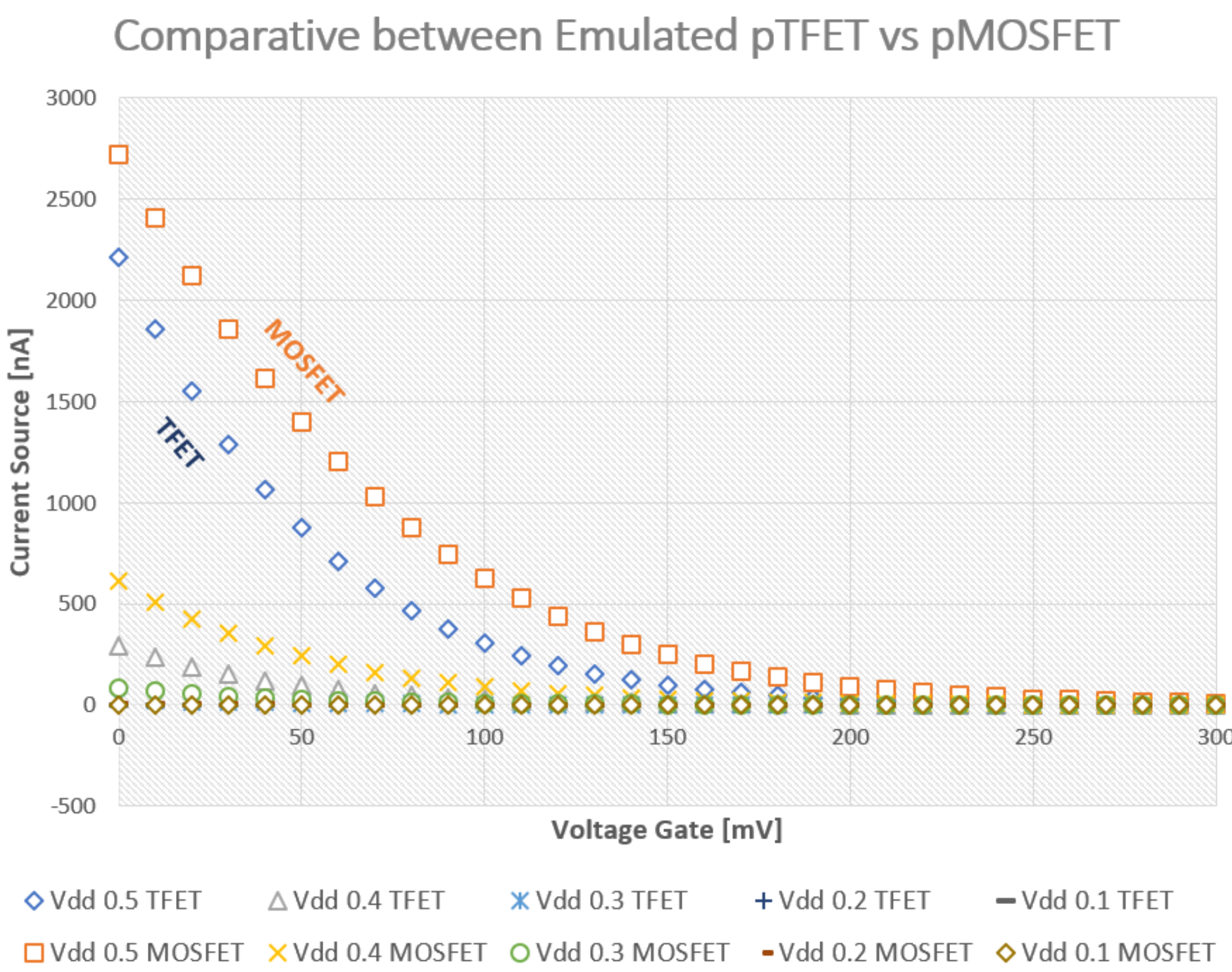


Figure 7: PMOSFET and PTFET Results.